How Do Our Models Stack Up?

Models A, B and C vs. GPT-3 and seven open small language models

Model comparison report — external baselines

November 2026

Abstract

We place three in-house language models in context:

- A: 54.5 M parameters, GPT-2 style, 0.38 B training tokens.
- B: 100.7 M parameters, Llama style, 3.42 B tokens.
- C: 343.7 M parameters, Llama style, 5.46 B tokens.

We compare them with the original GPT-3 family (Brown et al., 2020) and with seven openly released small models: GPT-2 124M and 355M, OPT-125M, Pythia-160M and 410M, and SmolLM2-135M and 360M. GPT-3 numbers are taken from the paper. Every other model, ours included, was run through one identical harness: the same raw text for bits per byte (bpb), the same benchmark items and prompts, and the same GPU.

The main findings:

- **C vs. GPT-3 Small.** C matches or beats GPT-3 Small on HellaSwag, ARC-Easy and PIQA, using 1/18 of its training compute and 55× fewer tokens.
- **B and C vs. GPT-2.** B and C model text better than the GPT-2 models of matching size, and beat them on ARC-Easy.
- **The modern models lead.** Models trained on far more tokens (Pythia: 300 B; SmolLM2: 2–4 T) beat C on almost every metric, and SmolLM2-360M is the strongest model tested. At this scale the training-token budget, not the architecture, separates the models.
- **A trails everything.** A is below every baseline on every metric.

1 The contenders

Model	Year	Family	Layers	d_{model}	Heads (KV)	Ctx	Positions / MLP	Vocab	Training data
GPT-3 Small / Medium†	2020	GPT	12 / 24	768 / 1024	12 / 16	2048	learned / GELU	50k	300B tok. (CC, WebText2, books, Wiki)
GPT-2 124M / 355M	2019	GPT	12 / 24	768 / 1024	12 / 16	1024	learned / GELU	50 257	WebText (≈40 GB)
OPT-125M	2022	GPT	12	768	12	2048	learned / ReLU	50 272	180B tok.
Pythia-160M / 410M	2023	GPT-NeoX	12 / 24	768 / 1024	12 / 16	2048	25% RoPE, parallel / GELU	50 304	300B tok. (The Pile)
SmolLM2-135M	2024	Llama	30	576	9 (3)	8192	RoPE / SwiGLU	49 152	2T tok.
SmolLM2-360M	2024	Llama	32	960	15 (5)	8192	RoPE / SwiGLU	49 152	4T tok.
A (ours)	2026	GPT	12	512	8	512	learned / GELU	32 000	0.38B tok. (37M unique ×10 epochs)
B (ours)	2026	Llama	12	768	12 (4)	4096	RoPE / SwiGLU	32 768	3.42B tok. + mid/post-training
C (ours)	2026	Llama	28	1024	16 (4)	4096	RoPE / SwiGLU	32 768	5.46B tok. + mid/post-training

Table 1: Architectures. Open-model values are read from each model's released `config.json`. †GPT-3 values are from Table 2.1 of the paper, and GPT-3 alternates dense and locally banded sparse attention. Training-token counts are as reported by each model's authors.

Two things to note about the line-up:

- **A and B are closest to the 125M class,** alongside GPT-2 124M, OPT-125M, Pythia-160M, SmolLM2-135M and GPT-3 Small.
- **C belongs to the 350M class,** alongside GPT-2 355M, Pythia-410M, SmolLM2-360M and GPT-3 Medium.

The defining difference is the training-token budget, which ranges over four orders of magnitude: from 0.4 B tokens for A to 4 T for SmolLM2-360M.

2 Method

- **Language-modelling quality: bits per byte (bpb).** Each model scores the same UTF-8 text with its own tokenizer:
  - four held-out domains (web, code, math, Wikipedia; about 0.55–0.6 MB each);
  - plus A's lower-cased test split ("A-test").

Each document starts from BOS and is scored in 512-token windows. bpb is the total negative log-likelihood in bits divided by the original byte count, so it is comparable across tokenizers.

- **Zero-shot benchmarks.** Each answer is scored by its log-likelihood:
  - HellaSwag and PIQA use length-normalised accuracy; ARC-Easy uses raw accuracy.
  - LAMBADA (the OpenAI version) requires an exact greedy prediction of the final word.
  - The items are the first 2 000 of each set, and all 1 838 for PIQA, identical for every measured model.
- **Speed.** We measure fp16 prefill throughput at batch 8 and sequence length 512 on one RTX 2070 Super.
- **GPT-3.** Its numbers are the published zero-shot values from Table H.1 of Brown et al. (2020), checked against the paper. They come from full test sets under the paper's own protocol, and were not re-measured here.
- **Harness check.** Our own checkpoints reproduce their training-log losses and benchmark scores. For example, C scores 35.7% on HellaSwag here and 35.75% in its own pipeline log.

3 Results

3.1 The full scoreboard

Model	Year	Params	Train tok.	Ctx	HellaSw.	ARC-E	PIQA	LAMB. acc	LAMB. ppl	bpb
GPT-3 Small†	2020	125M	300B	2048	33.7	43.6	64.6	42.7	18.6	–
GPT-3 Medium†	2020	356M	300B	2048	43.6	46.5	70.2	54.3	9.1	–
GPT-3 Large†	2020	760M	300B	2048	51.0	53.0	72.9	60.4	6.5	–
GPT-3 XL†	2020	1,320M	300B	2048	54.7	53.8	75.1	63.6	5.4	–
GPT-3 175B†	2020	174,600M	300B	2048	78.9	68.8	81.0	76.2	3.0	–
GPT-2 124M	2019	124M	n/r	1024	34.8	44.5	62.2	34.5	15.2	1.243
GPT-2 355M	2019	355M	n/r	1024	38.6	49.4	67.3	44.1	8.9	1.107
OPT 125M	2022	125M	180B	2048	34.5	43.2	61.8	39.4	10.9	1.194
Pythia 160M	2023	162M	300B	2048	33.7	44.3	61.8	34.8	12.8	1.024
Pythia 410M	2023	405M	300B	2048	38.2	52.8	68.2	51.2	5.5	0.885
SmolLM2 135M	2024	135M	2T	2048	38.9	64.3	68.6	43.8	7.8	0.895
SmolLM2 360M	2024	362M	4T	2048	43.5	70.3	72.7	54.4	5.0	0.796
A (this work)	2026	54M	0.38B	512	28.4	31.6	52.3	11.1	709.8	1.778
B (this work)	2026	101M	3B	4096	33.6	47.7	61.6	24.5	22.5	1.052
C (this work)	2026	344M	5B	4096	35.7	56.0	66.6	32.9	10.6	0.991

Table 2: Zero-shot results. All values are accuracy in percent, except LAMBADA perplexity and bpb (both lower is better). †GPT-3 rows are published values. Every other row was measured here on identical items. "Train tok." is the pre-training token count reported by the authors (n/r = not reported).

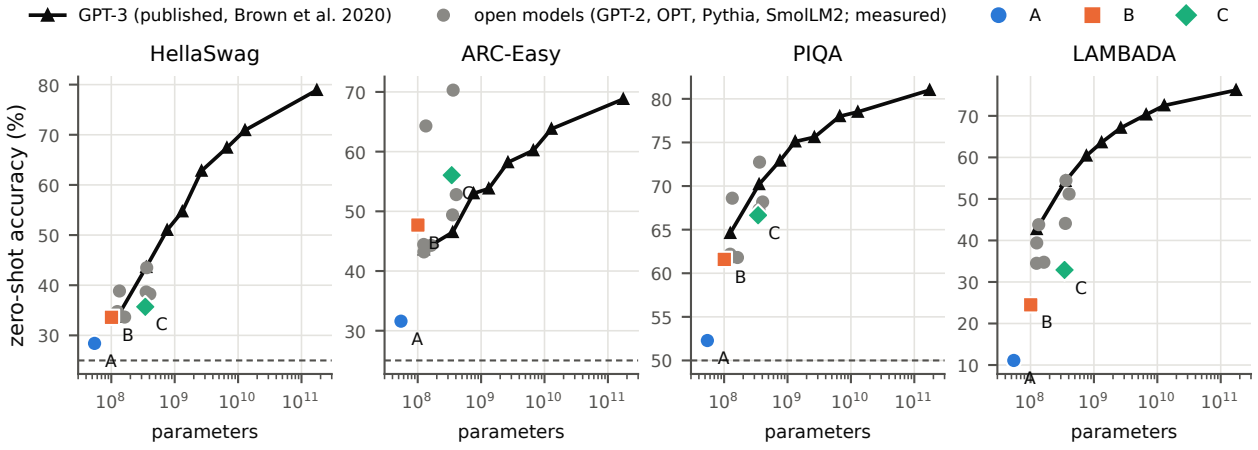


Figure 1: Accuracy vs. parameter count. The black line is the published GPT-3 family. Grey points are the open models, measured here. Coloured markers are our models. Dashed lines mark chance.

The GPT-3 curve in Figure 1 is a useful yardstick: it shows what 300 B tokens buy at each model size.

- **On ARC-Easy,** C and B sit *above* the GPT-3 curve, and both SmolLM2 models sit far above it. This reflects the educational, science, code and math content of modern data mixes.
- **On HellaSwag and PIQA,** C lands near GPT-3 Small, a model about a third of its size.
- **On LAMBADA,** all three of our models are well below the curve.

3.2 Head-to-head: C against the 350M class and GPT-3

	HellaSwag	ARC-Easy	PIQA	LAMBADA	bpb ↓	Train tokens
GPT-3 Small (125M)†	33.7	43.6	64.6	42.7	–	300B
GPT-3 Medium (350M)†	43.6	46.5	70.2	54.3	–	300B
GPT-2 355M	38.6	49.4	67.3	44.1	1.107	n/r
Pythia-410M	38.2	52.8	68.2	51.2	0.885	300B
SmolLM2-360M	**43.5**	**70.3**	**72.7**	**54.4**	**0.796**	4T
C (ours)	35.7	56.0	66.6	32.9	0.991	5.46B

Table 3: C against models of the same size, plus GPT-3 Small. †GPT-3 values are published.

What Table 3 shows for **C**:

- **Ahead of GPT-2 355M,** the closest model of the same size and era-defining architecture, on bpb (0.991 vs. 1.107) and on ARC-Easy.
- **Behind GPT-2 355M** on HellaSwag, PIQA and especially LAMBADA.
- **Behind Pythia-410M and SmolLM2-360M** on every metric except Pythia's ARC-Easy. Those models saw 55× and 730× more tokens.

3.3 Language-modelling quality across domains

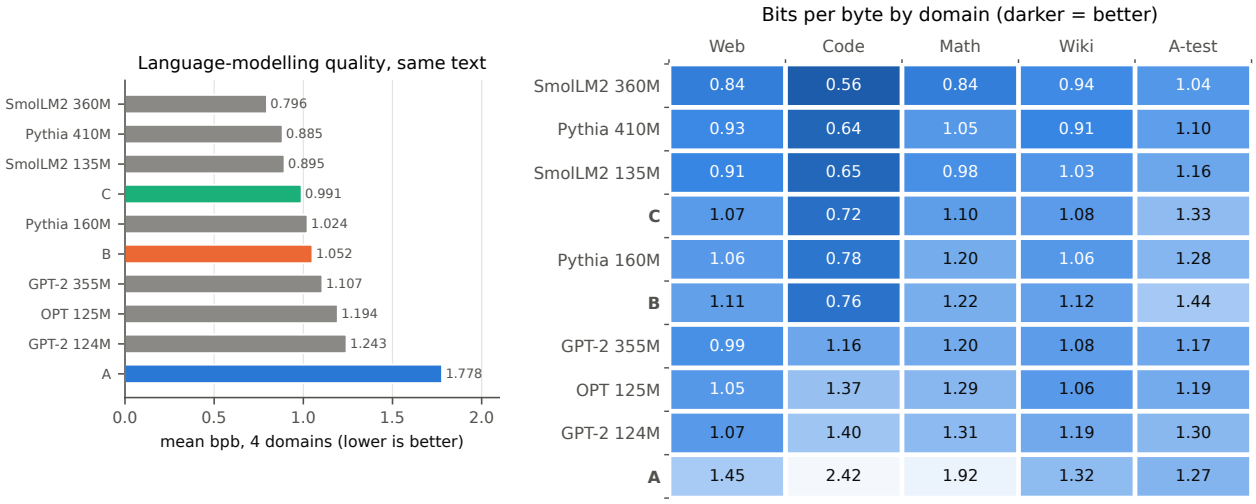


Figure 2: Left: mean bpb over the four domains, ranked. Right: bpb per domain, with rows sorted by the mean. Bold rows are our models.

Figure 2 shows how the models rank on raw text:

- **Overall,** **C** ranks 4th of 10 and **B** 6th, ahead of all three 2019–2022 models (GPT-2 124M and 355M, OPT-125M).
- **On code,** our models are much stronger than GPT-2 and OPT, whose training data contained almost no source code.
- **On Wikipedia and A-test,** Pythia and GPT-2 355M do relatively well, because their corpora contain more encyclopedic and news text.
- **A** is last on every domain except its own A-test, where it beats B, C, GPT-2 124M and (narrowly) Pythia-160M.

3.4 Data and compute efficiency

Figure 3 shows how efficiently each model turns data and compute into accuracy:

- **Per token of data,** B and C are by far the most efficient models in the comparison: they reach 33–36% HellaSwag with 3–5 B tokens, while the baselines needed 180 B–4 T tokens to reach about 34–44%.
- **Per FLOP,** C reaches GPT-3 Small's HellaSwag score with about 5% of its compute.
- **The gain is not free.** Accuracy keeps rising with tokens, so the extra data of the modern models still buys a large margin.

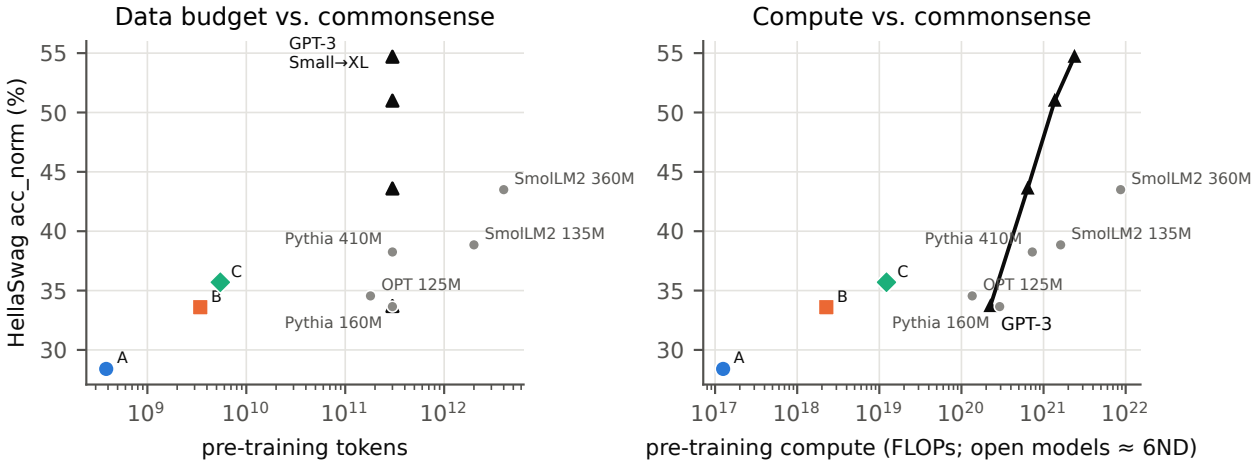


Figure 3: HellaSwag vs. pre-training tokens (left) and vs. pre-training compute (right). Compute for open models is estimated as $6ND$; GPT-3 compute is from Table D.1 of the paper. GPT-2 is omitted because its token count was not published.

3.5 Speed vs. quality

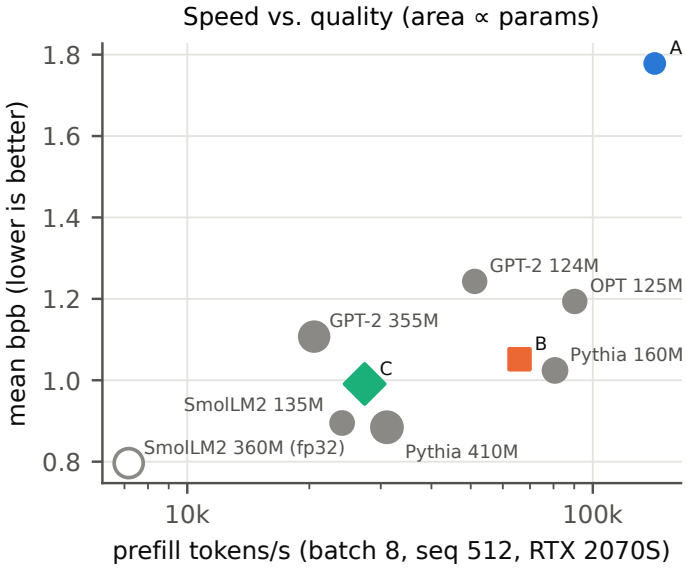


Figure 4: Prefill throughput vs. mean bpb on one RTX 2070 Super (fp16, batch 8, sequence length 512). Marker area is proportional to parameters. SmolLM2-360M (hollow marker) had to run in fp32 because its activations overflow fp16, so its speed is not comparable.

On speed vs. quality (Figure 4):

- **B** sits just behind Pythia-160M, which is both slightly faster and slightly better in bpb.
- **SmolLM2-135M** is deep and narrow (30 layers, d=576), so it is slow for its size.
- **C** is about as fast as Pythia-410M and SmolLM2-135M.
- **A** is the fastest model, but far behind on quality.

4 Verdict per model

A **(54M).** Below every baseline on every metric. It is data-starved: about 7 tokens per parameter, repeated over 10 epochs. It also lower-cases its input and is limited to 512 tokens. Its closest peers, GPT-2 124M and OPT-125M, are 2–3 years older designs, yet clearly better because they saw 100–500× more data. Value: an educational baseline.

B **(101M).** Comparable to GPT-2 124M and OPT-125M on commonsense (HellaSwag 33.6 vs. 34.8/34.5). Better on ARC-Easy and bpb, and weaker on LAMBADA. Unlike any of the baselines, it is instruction-tuned and chat-capable.

C (344M). Roughly the level of GPT-3 Small, and of GPT-2 355M on bpb and ARC-Easy. It is the most data-efficient model in the comparison. With B, it is one of only two instruction-tuned, chat-capable models here, and with B and SmolLM2, one of the few with a context window of at least 4k tokens. It is clearly behind the 2023–24 models trained on 300 B–4 T tokens, with LAMBADA as its weakest point.

Recommendation. For C, the cheapest path to a better model is more pre-training tokens, not a larger network. Pythia-410M, a similar size trained on 55× more tokens, gains about 18 LAMBADA points and 0.1 bpb over C. C's own LAMBADA score was still rising steeply when pre-training stopped.

5 Caveats

- **GPT-3 protocol.** GPT-3 numbers come from the paper's protocol (full test sets; ARC uses unconditional-probability normalisation), so they are not strictly comparable with the measured rows. Our 2 000-item subsets carry about ±2 points of sampling noise.
- **Test-text distribution.** The bpb text comes from the data sources of the B/C pipeline. This favours models trained on similar web, code and math mixes (B, C, SmolLM2), and penalises GPT-2, OPT and Pythia on code.
- **Base vs. chat models.** The baselines are base models, while B and C are post-trained chat models scored here without a chat template. Post-training changed C's benchmark scores by under 2 points, but it does slightly raise bpb.
- **Speed.** Speed is measured on consumer hardware without fused kernels. Relative order matters more than absolute numbers.

Reproducing

```
python3 eval_compare.py --model <A|B|C|gpt2|gpt2-medium|opt-125m|pythia-160m|
                                 pythia-410m|smollm2-135m|smollm2-360m> --gpu 0
python3 make_figs.py && python3 make_figs_baselines.py
latexmk -pdf report_baselines.tex
```

Reference. T. Brown et al., *Language Models are Few-Shot Learners*, NeurIPS 2020, arXiv:2005.14165 (Tables 2.1, D.1, H.1).